AMCS 206 — Midterm Cheat Sheet

Taylor Series · Iterative & Direct Solvers · Least Squares

1. Taylor Series & Error

Approximates $f(x)$ near expansion point x_0 by a polynomial.

$$f(x) \approx P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

Lagrange remainder (exact error): for some ξ between x_0 and x,

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$$

Error bound:

$$|R_n(x)| \le \frac{M_{n+1}}{(n+1)!}|x-x_0|^{n+1}, \quad M_{n+1} = \max_{\xi} |f^{(n+1)}(\xi)|$$

Common expansions ($x_0 = 0$): $e^x = \sum \frac{x^k}{k!}$, $\sin x = \sum \frac{(-1)^k x^{2k+1}}{(2k+1)!}$, $\cos x = \sum \frac{(-1)^k x^{2k}}{(2k)!}$, $\frac{1}{1-x} = \sum x^k$.

Pitfalls: forgetting $(n+1)!$; missing absolute value; using $\xi = x_0$ instead of bounding $|f^{(n+1)}|$ over the whole interval.

2. Iterative Solvers for $Ax = b$

Write $A = M - N$ (splitting). Iteration:

$$Mx^{(k+1)} = Nx^{(k)} + b \iff x^{(k+1)} = x^{(k)} + M^{-1}r^{(k)}$$

with residual $r^{(k)} = b - Ax^{(k)}$ and iteration matrix $T = I - M^{-1}A = M^{-1}N$.

Convergence: $x^{(k)} \to x$ for any $x^{(0)}$ **iff** $\rho(T) < 1$, where $\rho(T) = \max_i |\lambda_i(T)|$ is the spectral radius. Smaller $\rho \Rightarrow$ faster convergence; asymptotic rate $\approx -\log_{10} \rho(T)$ digits per step.

Sufficient condition: if A is strictly diagonally dominant, Jacobi and Gauss–Seidel both converge. If A is SPD, Gauss–Seidel and SOR ($0 < \omega < 2$) converge.

Decompose $A = D + L_{\text{off}} + U_{\text{off}}$ (diagonal, strict lower, strict upper).

Method	M	Notes
Jacobi	D	parallel; slow
G–S	$D + L_{\text{off}}$	sequential; faster
SOR	$\frac{1}{\omega}D + L_{\text{off}}$	$0 < \omega < 2$; accelerates

Componentwise (Jacobi):

$$x_i^{(k+1)} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j \ne i} a_{ij}x_j^{(k)}\Big)$$

Gauss–Seidel: same, but use updated $x_j^{(k+1)}$ for $j < i$.

Pitfall: eigenvalues of a triangular matrix = its diagonal entries. Use this when computing $\rho(T)$ for Gauss–Seidel on small systems.

3. Direct Methods & Factorization

LU: $A = LU$ (L unit lower-tri, U upper-tri). Solve $Ax = b$ by forward sub $Ly = b$, then back sub $Ux = y$. Cost $\sim \frac{2}{3}n^3$.

Partial pivoting ($PA = LU$): at step k swap rows so $|a_{kk}|$ is the largest in its column below. Prevents 0-pivots, limits round-off growth.

Scaled partial pivoting: compare $|a_{ik}|/s_i$ where $s_i = \max_j |a_{ij}|$ is the row's scale factor; choose row maximizing this ratio.

Cholesky: if A is SPD then $A = LL^\top$ (or $LDL^\top$, D diagonal > 0). *No pivoting needed.* Cost $\sim \frac{1}{3}n^3$ — half of LU. Existence of $LL^\top$ with real positive diagonal $\Leftrightarrow$ A SPD.

Condition number: $\kappa(A) = \|A\|\,\|A^{-1}\|$. Relative error satisfies $\frac{\|\delta x\|}{\|x\|} \lesssim \kappa(A)\frac{\|\delta b\|}{\|b\|}$.

Check SPD: symmetric *and* all leading principal minors > 0 (or all eigenvalues > 0, or $x^\top Ax > 0\ \forall x \ne 0$).

4. Least Squares (overdetermined $Ax \approx b$)

For $A \in \mathbb{R}^{m\times n}$, $m > n$, minimize $\|b - Ax\|_2^2$.

Normal equations:

$$A^\top A\tilde{x} = A^\top b$$

Unique solution iff A has full column rank. The residual $r = b - A\tilde{x}$ is orthogonal to $\mathrm{Col}(A)$: $A^\top r = 0$.

Polynomial fit $y \approx c_0 + c_1x + \cdots + c_mx^m$: use Vandermonde matrix

$$A = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_N & x_N^2 & \cdots & x_N^m \end{pmatrix}, \quad b = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}$$

Linearizable models: $y = ae^{bx} \Rightarrow \ln y = \ln a + bx$; $y = ax^b \Rightarrow \ln y = \ln a + b\ln x$.

Error metrics:

$$\mathrm{SSE} = \sum r_i^2, \quad \mathrm{RMS} = \sqrt{\tfrac{1}{N}\sum r_i^2}, \quad R^2 = 1 - \tfrac{\mathrm{SSE}}{\sum(y_i - \bar{y})^2}$$

Pitfall: $\kappa(A^\top A) = \kappa(A)^2$. Normal equations lose *twice* the digits. Prefer QR ($A = QR \Rightarrow Rx = Q^\top b$) for ill-conditioned problems.

5. Quick Reference

Quantity	Formula
Residual	$r = b - Ax$
Spectral radius	$\rho(T) = \max\|\lambda_i(T)\|$
Jacobi T	$-D^{-1}(L_{\text{off}} + U_{\text{off}})$
G–S T	$-(D + L_{\text{off}})^{-1}U_{\text{off}}$
LU cost	$\frac{2}{3}n^3$
Cholesky cost	$\frac{1}{3}n^3$
Normal eqns	$A^\top A\tilde{x} = A^\top b$
$\kappa_2(A)$	$\sigma_{\max}/\sigma_{\min}$

6. Taylor: bounding the error

Example. Bound error of $P_3(x)$ for $f(x) = e^x$ about $x_0 = 0$ on $[0, 0.5]$.
$f^{(4)}(\xi) = e^\xi$, so $M_4 = \max_{[0,0.5]} e^\xi = e^{0.5} \approx 1.6487$.

$$|R_3(0.5)| \le \frac{1.6487}{4!}(0.5)^4 = \frac{1.6487 \cdot 0.0625}{24} \approx 4.29 \times 10^{-3}.$$

Recipe: (1) write R_n; (2) bound $|f^{(n+1)}|$ on *whole* interval, not at x_0; (3) substitute largest $|x - x_0|$.

7. Jacobi / G–S by hand

For 2×2 system $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} x = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$, start $x^{(0)} = 0$:

$$x_1^{(k+1)} = \tfrac{5 - x_2^{(k)}}{4}, \qquad x_2^{(k+1)} = \tfrac{5 - 2x_1^{(\cdot)}}{3}$$

Jacobi uses $x_1^{(k)}$; Gauss–Seidel uses the *just-updated* $x_1^{(k+1)}$.

Spectral radius check (Jacobi): $T_J = -D^{-1}(L_{\text{off}} + U_{\text{off}}) = \begin{pmatrix} 0 & -1/4 \\ -2/3 & 0 \end{pmatrix}$, $\lambda^2 = \frac{1}{6} \Rightarrow \rho = 1/\sqrt{6} \approx 0.408 < 1$ ✓.

8. LU with partial pivoting

Algorithm (step k):

1. Find row $p \ge k$ maximizing $|a_{pk}|$.
2. Swap rows $k \leftrightarrow p$ in A and in P; swap already-computed L entries in columns $< k$.
3. Eliminate: $\ell_{ik} = a_{ik}/a_{kk}$, $a_{ij} \leftarrow a_{ij} - \ell_{ik}a_{kj}$ for $i > k$.

Result: $PA = LU$. Solve $Ly = Pb$, then $Ux = y$.

Scaled variant: compute $s_i = \max_j |a_{ij}|$ *once* at start; at step k pick row maximizing $|a_{ik}|/s_i$. Do *not* rescale the actual rows.

9. Cholesky by hand

For SPD A, $A = LL^\top$ with

$$\ell_{jj} = \sqrt{a_{jj} - \textstyle\sum_{k<j} \ell_{jk}^2}$$

$$\ell_{ij} = \tfrac{1}{\ell_{jj}}\Big(a_{ij} - \textstyle\sum_{k<j} \ell_{ik}\ell_{jk}\Big), \ i > j$$

If any $\ell_{jj}^2 \le 0$ appears, A is *not* SPD — stop.

10. Least squares: fitting a line

Fit $y = c_0 + c_1x$ to data (x_i, y_i), $i = 1, \ldots, N$. Normal equations reduce to:

$$\begin{pmatrix} N & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}\begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} \sum y_i \\ \sum x_iy_i \end{pmatrix}$$

Closed form:

$$c_1 = \frac{N\sum x_iy_i - \sum x_i \sum y_i}{N\sum x_i^2 - (\sum x_i)^2}, \quad c_0 = \bar{y} - c_1\bar{x}.$$

Exponential fit $y = ae^{bx}$: take $Y = \ln y$, solve linear LS for $(\ln a, b)$, then $a = e^{\ln a}$. *Note:* this minimizes error in $\ln y$, not y.

RMS vs SSE: $\mathrm{SSE} = \sum r_i^2$; $\mathrm{RMS} = \sqrt{\mathrm{SSE}/N}$; sometimes $\sqrt{\mathrm{SSE}/(N-n)}$ (unbiased, n = # parameters) — check convention.

11. When to use what

Situation	Best tool
Small dense $Ax = b$	LU + partial pivoting
SPD dense	Cholesky ($LL^\top$)
Large sparse, SPD	Gauss–Seidel, SOR, CG
Large sparse, general	GMRES / BiCGStab
Ill-cond. least squares	QR factorization
Well-cond. least squares	Normal equations
Smooth f, need value/deriv	Taylor about nearby x_0

12. Final checklist

- Residual $r = b - Ax$, *not* $Ax - b$ (sign matters in updates).
- Check $\rho(T) < 1$ *before* iterating; diag. dominance is sufficient, not necessary.
- For SPD tests: symmetric *and* positive diagonal is necessary but not sufficient — check minors/eigs.
- In Taylor bounds, bound $|f^{(n+1)}|$ over the *whole* interval.
- Normal-equation condition number is squared — watch precision loss.
- Triangular matrix $\Rightarrow$ eigenvalues = diagonal entries.
- Verify pivoting: after LU, check $PA = LU$ holds.